

SOHAR MAJIS, Oman

WMO: 412460

Lat: 24.4674N

Lon: 56.6445E

Elev: 2

StdP: 101.30

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.4	14.7	1.2	4.1	26.8	3.6	4.9	25.1	9.0	22.4	7.9	22.9	2.4	240	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.2	40.4	23.5	38.2	26.1	37.0	27.4	32.1	35.3	31.4	34.8	30.9	34.3	4.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.4	29.6	34.6	30.6	28.3	34.1	30.0	27.3	33.7	112.1	35.4	108.3	34.8	105.2	34.4	35.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.5	6.0	DB	11.3	45.0	2.0	1.8	9.9	46.3	8.7	47.4	7.6	48.4	6.2	49.7	(4)
(5)			WB	7.1	32.5	1.7	1.4	5.9	33.5	4.9	34.3	3.9	35.1	2.7	36.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	20.1	20.9	23.4	27.7	31.8	33.5	33.3	32.3	30.9	28.8	24.9	21.7	(6)
(7)		DBStd	5.22	1.77	2.04	2.25	2.66	2.48	1.96	1.48	1.62	1.45	1.88	1.88	1.74	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	8	4	3	0	0	0	0	0	0	0	0	0	1	(9)
(10)		CDD10.0	6382	312	306	417	532	677	704	724	692	628	583	446	362	(10)
(11)		CDD18.3	3349	58	76	159	282	418	454	466	434	378	325	196	104	(11)
(12)		CDH23.3	44805	194	323	1172	3390	6453	7346	7401	6569	5457	4194	1761	546	(12)
(13)	CDH26.7	25559	9	25	234	1532	4044	4948	4921	4092	3103	2106	503	42	(13)	
(14)	Wind	WSAvg	2.6	2.6	2.6	2.6	2.6	2.6	2.8	2.8	2.5	2.3	2.4	2.4	(14)	
(15)	Precipitation	PrecAvg	93	12	20	21	8	1	1	2	1	5	7	11	(15)	
(16)		PrecMax	313	74	91	80	32	3	5	55	6	5	112	55	169	(16)
(17)		PrecMin	25	0	6	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	65	18	21	25	9	1	1	10	1	1	21	11	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.8	29.2	33.4	39.1	42.9	44.0	41.1	40.2	37.9	36.2	32.2	29.5	(19)
(20)			MCWB	18.2	18.9	19.6	20.0	21.0	21.3	26.0	26.3	25.3	23.7	22.6	18.9	(20)
(21)		2%	DB	26.0	27.1	30.3	35.8	39.8	40.5	38.1	37.3	35.7	34.8	31.0	27.7	(21)
(22)			MCWB	19.4	20.2	20.5	21.3	21.2	24.1	28.2	28.6	27.2	25.1	23.5	20.5	(22)
(23)		5%	DB	25.1	26.0	28.9	33.9	37.3	38.0	36.8	36.0	34.8	33.9	30.2	26.8	(23)
(24)			MCWB	19.4	20.2	21.3	23.3	24.8	27.6	29.2	29.2	28.1	25.7	23.4	20.5	(24)
(25)		10%	DB	24.4	25.2	27.7	32.4	36.0	36.8	35.9	35.0	34.0	33.0	29.4	26.1	(25)
(26)			MCWB	18.9	20.0	21.3	23.5	25.9	28.2	29.5	29.4	28.3	25.9	23.0	20.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.4	23.5	25.3	28.5	32.2	33.9	33.0	32.4	31.1	29.9	26.3	23.9	(27)
(28)			MCD B	25.3	26.1	28.4	32.3	35.6	36.6	35.3	35.9	34.3	33.2	30.1	26.6	(28)
(29)		2%	WB	21.4	22.7	24.2	27.2	30.3	32.2	31.9	31.6	30.2	28.7	25.3	22.6	(29)
(30)			MCD B	24.6	25.5	27.3	31.5	34.4	35.1	35.0	35.1	33.5	32.5	29.6	26.1	(30)
(31)		5%	WB	20.6	22.0	23.5	26.4	29.4	31.1	31.3	31.0	29.6	27.9	24.6	21.8	(31)
(32)			MCD B	24.1	24.9	26.8	30.6	33.9	34.8	34.6	34.1	33.0	31.9	29.1	25.6	(32)
(33)		10%	WB	19.8	21.2	22.8	25.5	28.7	30.5	30.9	30.3	29.2	27.1	23.8	20.9	(33)
(34)			MCD B	23.6	24.3	26.2	29.9	33.4	34.5	34.3	33.6	32.6	31.5	28.4	24.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	8.2	7.9	8.0	7.7	6.2	4.5	4.8	6.1	8.2	8.4	8.3	(35)
(36)			MCDBR	8.6	8.7	9.2	9.9	9.6	8.8	6.6	6.9	7.3	8.7	8.9	9.0	(36)
(37)			MCWBR	5.4	5.9	7.2	7.7	8.5	8.4	6.0	5.3	5.8	7.6	5.9	5.6	(37)
(38)		5% WB	MCD BR	8.0	7.1	7.0	7.9	7.3	6.5	5.0	5.2	6.0	7.3	8.4	8.1	(38)
(39)			MCWBR	5.4	5.3	6.0	7.6	8.0	6.6	4.5	4.0	4.8	6.6	5.9	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.437	0.492	0.551	0.588	0.594	0.659	0.739	0.694	0.595	0.502	0.452	0.435	(40)	
(41)		taud	2.169	2.001	1.847	1.792	1.777	1.678	1.587	1.661	1.815	2.004	2.151	2.171	(41)	
(42)		Ebn at Noon	808	788	762	743	736	684	632	659	716	767	786	791	(42)	
(43)		Edh at Noon	134	169	207	223	225	247	271	251	210	166	135	129	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.33	5.07	5.96	6.74	7.48	7.51	6.97	6.69	6.30	5.59	4.60	4.10	(44)	
(45)		RadStd	0.26	0.33	0.37	0.46	0.21	0.26	0.26	0.20	0.13	0.14	0.11	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.68	+1.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.40	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	+128	+126

Nomenclature: See separate page